QUEUING THEORY: PART 3

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Question: Customers arrive at a counter randomly every 4 minutes on an average. Determine the probability that the inter-arrival time will be from 1 minute to 2 minutes. If the service time distribution is also exponential with expected time of 3 minutes, find out (i) the percentage utilization of the service facility, (ii) length of system and (iii) length of queue. What is the probability that there will be two customers in the system at any point in time?

Solution: Given,

Mean inter-arrival time = 4 minutes Hence, Arrival rate, $\lambda=1/4$ customers/minute

Mean service time = 3 minutes Hence, Service rate, $\mu=1/3$ customers/minute

Both arrival and service times follow an exponential distribution

Probability that inter-arrival time is from t_1 = 1 to t_2 = 2 minutes

For exponential distribution:

$$P(t_1 \le T \le t_2) = e^{-\lambda t_1} - e^{-\lambda t_2}$$

$$= e^{-1/4 \times 1} - e^{-1/4 \times 2}$$

$$= 0.1723.$$

Remember: $\mathrm{P}\{x \le A\} = 1 - e^{-\lambda A}$.

Hence, the probability is 17.23%.

(i) Utilization,

$$\rho = \frac{\lambda}{\mu} = \frac{1/4}{1/3} = 0.75$$

Hence, utilization of the service facility = 75%.

(ii) Length of system,

$$L = \frac{\rho}{1-\rho} = \frac{0.75}{1-0.75} = 3$$

Hence, length of the system is 3 customers.

(iii) Length of queue,

$$L_q = L_s - \rho = \frac{\rho^2}{1-\rho} = \frac{0.75^2}{1-0.75} = 2.25$$

Hence, length of the queue = 2.25 customers

Queue (Waiting Line)	System (Queue + Service)
Customers who have arrived but are **not being served yet.**	Customers **waiting** in line including the customer being served.

Probability that there are n customers in the system is given by

$$P(n) = \rho^n (1-\rho)$$

For n = 2,

$$P(2) = \rho^2 (1-\rho) = 0.75^2 (1-0.75) = 0.1406$$

Hence, probability that there will be 2 customers in the system = 14.06%.

- **Example:** A coffee shop owner runs three counters for providing coffee to customers. Depending on the number of customers, he manages the number of counters with his workers as shown in table. Customers arrive at a mean rate of 8 customers per hour. Check-out time per customer is exponential with mean 10 minutes. Determine the probability that one counter will be in operation.

Number of customers	Number of counters in operation
1 to 3	1
4 to 5	2
More than 5	3

- **Solution:**

$\lambda_n = \lambda = 8$ customers per hour, $n = 0, 1, 2, \ldots.$

$$\mu_n = \begin{cases} \dfrac{60}{10} = 6 \text{ customers per hour}, & n = 0, 1, 2, 3 \\ 2 \times 6 = 12 \text{ customers per hour}, & n = 4, 5 \\ 3 \times 6 = 18 \text{ customers per hour}, & n = 6, 7, \ldots.. \end{cases}$$

Thus

$$p_1 = \left(\frac{8}{6}\right) p_0 = 1.33\, p_0$$

$$p_2 = \left(\frac{8}{6}\right)^2 p_0 = 1.33^2\, p_0$$

$$p_3 = \left(\frac{8}{6}\right)^3 p_0 = 1.33^3\, p_0$$

$$p_4 = \left(\frac{8}{6}\right)^3 \left(\frac{8}{12}\right) p_0 = \left(1.33^3 \times 0.66\right) p_0$$

$$p_5 = \left(\frac{8}{6}\right)^3 \left(\frac{8}{12}\right)^2 p_0 = \left(1.33^3 \times 0.66^2\right) p_0$$

$$p_{n\geq 6} = \left(\frac{8}{6}\right)^3 \left(\frac{8}{12}\right)^2 \left(\frac{8}{18}\right)^{n-5} p_0 = \left\{\left(1.33^3 \times 0.66^2\right) \times \left(0.44\right)^{n-5}\right\} p_0$$

The value of p_0 is determined from

$$p_0 + 1.33\, p_0 + 1.33^2\, p_0 + 1.33^3\, p_0 + \left(1.33^3 \times 0.66\right) p_0$$
$$+ \left(1.33^3 \times 0.66^2\right) p_0 + \left\{\left(1.33^3 \times 0.66^2\right) \times 0.44^{n-5}\right\} p_0 + ... = 1$$

or

$$p_0 + p_0 \left\{ \begin{array}{l} 1.33 + 1.33^2 + 1.33^3 + \left(1.33^3 \times 0.66\right) + \left(1.33^3 \times 0.66^2\right) + \left(1.33^3 \times 0.66^2 \times 0.44\right) \\ + \left(1.33^3 \times 0.66^2 \times 0.44^2\right) + \left(1.33^3 \times 0.66^2 \times 0.44^3\right) + \left(1.33^3 \times 0.66^2 \times 0.44^4\right) + ... \end{array} \right\} = 1$$

or

$$p_0 + p_0 \left[\begin{array}{l} \left\{1.33 + 1.33^2 + 1.33^3 + \left(1.33^3 \times 0.66\right)\right\} \\ + \left\{\left(1.33^3 \times 0.66^2\right)\right\}\left\{1 + 0.44 + 0.44^2 + 0.44^3 + 0.44^4 + ...\right\} \end{array} \right] = 1$$

Using the geometric sum series for infinite terms

$$\sum_{i=0}^{\infty} x^{\,i} = \frac{1}{1-x}, \quad |x| < 1$$

Solving, we obtain

$$p_0 + p_0\left[7.004 + \left\{\left(1.33^3 \times 0.66^2\right)\right\}\left\{1 + 0.44 + 0.44^2 + 0.44^3 + 0.44^4 + ...\right\}\right] = 1$$

$$\Rightarrow p_0 + p_0\left[7.004 + 0.771 \times \left(\frac{1}{1-0.44}\right)\right] = 1$$

$$\Rightarrow 9.381 p_0 = 1$$

$$\Rightarrow p_0 = 0.107$$

The probability that one counter will be opened will be calculated by assuming that there will be at most 3 customers in the coffee shop is

$$p_0 + p_1 + p_2 + p_3 = p_0 + 1.33 p_0 + 1.33^2 p_0 + 1.33^3 p_0$$

$$= p_0(1 + 1.33 + 1.33^2 + 1.33^3)$$

$$= 0.107 \times 6.452 = 0.69$$

Specialized Poisson Queues

- **Specialized Poisson Queues** refer to queuing models that are **extensions or variations** of the basic Poisson queuing systems.
- These are designed to handle **special real-world features** such as priorities, bulk arrivals, finite capacity, or customer behaviors like balking and reneging.
- This model introduce new features like Priorities, Group arrivals, Customer impatience, Feedback or retries and State-dependent behavior

Specialized Poisson Queues

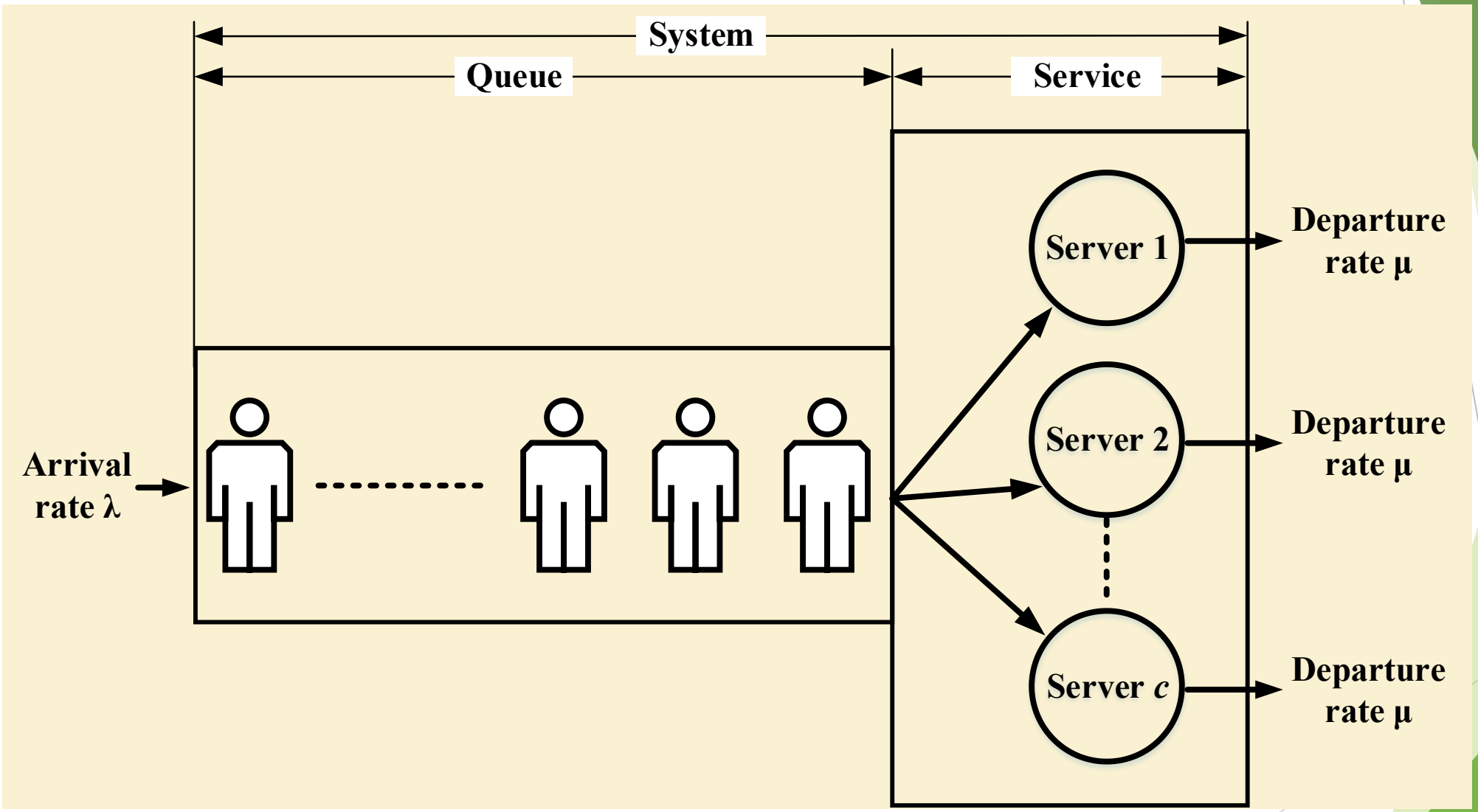


Figure Representation of queuing system with ***c*** parallel servers

Specialized Poisson Queues

Figure in last slide depicts the Poisson queuing situation with *c* parallel servers. The arrival rate at the system is λ customers per unit time. All parallel servers are identical. The service rate for any server is μ customers per unit time.

The characteristics of the queuing situation is

$$(a/b/c):(d/e/f)$$

where

a = Arrivals distribution

b = Departures (service time) distribution

c = Number of parallel servers (= 1, 2, ..., ∞)

d = Queue discipline

e = Maximum queue size (finite or infinite) allowed in the system (in-queue plus in-service)

f = Size of the calling source (finite or infinite)

Specialized Poisson Queues

The standard notation for representing the arrivals and departures distributions (symbols *a* and *b*) is

M = Markovian (or Poisson) arrivals or departures distribution (or equivalently exponential inter-arrival or service time distribution)

D = Constant (deterministic) time

E_k = Erlang or gamma distribution of time (or, equivalently, the sum of independent exponential distributions)

GI = General (generic) distribution of inter-arrival time

G = General (generic) distribution of service time

Specialized Poisson Queues

The queue discipline notation (symbol *d*) includes:

FIFO = First-in, first-out

LIFO = Last-in, first-out

SIRO = Service in random order

GD = General discipline (i.e., any type of discipline)

For illustration of the notation, the model (M/D/5):(GD/15/∞) uses Markovian (Poisson) arrivals, constant service time and 5 parallel servers. The queue discipline is **GD** and there is a limit of 15 customers on the entire system. The size of the source from which customer arrive is infinite.

Steady State Measurement of Performance

Commonly used measures of performance in a queuing situation are

L_s = Expected number of customers in *system*

L_q = Expected number of customers in *queue*

W_s = Expected waiting time in *system*

W_q = Expected waiting time in *queue*

$\bar{c}$ = Expected number of busy servers

These measures are derived from steady-state probability of n in the system p_n as

$$L_s = \sum_{n=1}^{\infty} n p_n$$

$$L_q = \sum_{n=\bar{c}+1}^{\infty} (n - \bar{c}) p_n$$

Steady State Measurement of Performance

The relationship between L_s and W_s (also L_q and W_q) is referred to as **Little's formula:**

$$L_s = \lambda_{\text{eff}} W_s$$

$$L_q = \lambda_{\text{eff}} W_q$$

The parameter λ_{eff} is the effective arrival rate at the system. It is equal to the nominal arrival rate λ when all arriving customers can join the system.

The relationship between W_s and W_q is given by

$$\begin{pmatrix}\text{Expected waiting time}\\ \text{in system, } W_s\end{pmatrix} = \begin{pmatrix}\text{Expected waiting time}\\ \text{in queue, } W_q\end{pmatrix} + (\text{Expected service time})$$

Steady State Measurement of Performance

or

$$W_s = W_q + \frac{1}{\mu}$$

Multiplying both sides with λ_{eff}

$$\lambda_{eff} W_s = \lambda_{eff} W_q + \frac{\lambda_{eff}}{\mu}$$

$$L_s = L_q + \frac{\lambda_{eff}}{\mu}$$

The difference between the average number in the system, L_s and the average number in the queue, L_q must equal the average number of busy servers, $\bar{c}$. Thus,

$$\bar{c} = L_s - L_q = \frac{\lambda_{eff}}{\mu}$$

$$\left(\text{Facility utilization}\right) = \frac{\bar{c}}{c}$$